

Final Report

Task 9: Wireshark – Capture & Analyze Packets

STEP 1 — Install Wireshark

```
sudo apt install wireshark -y
```

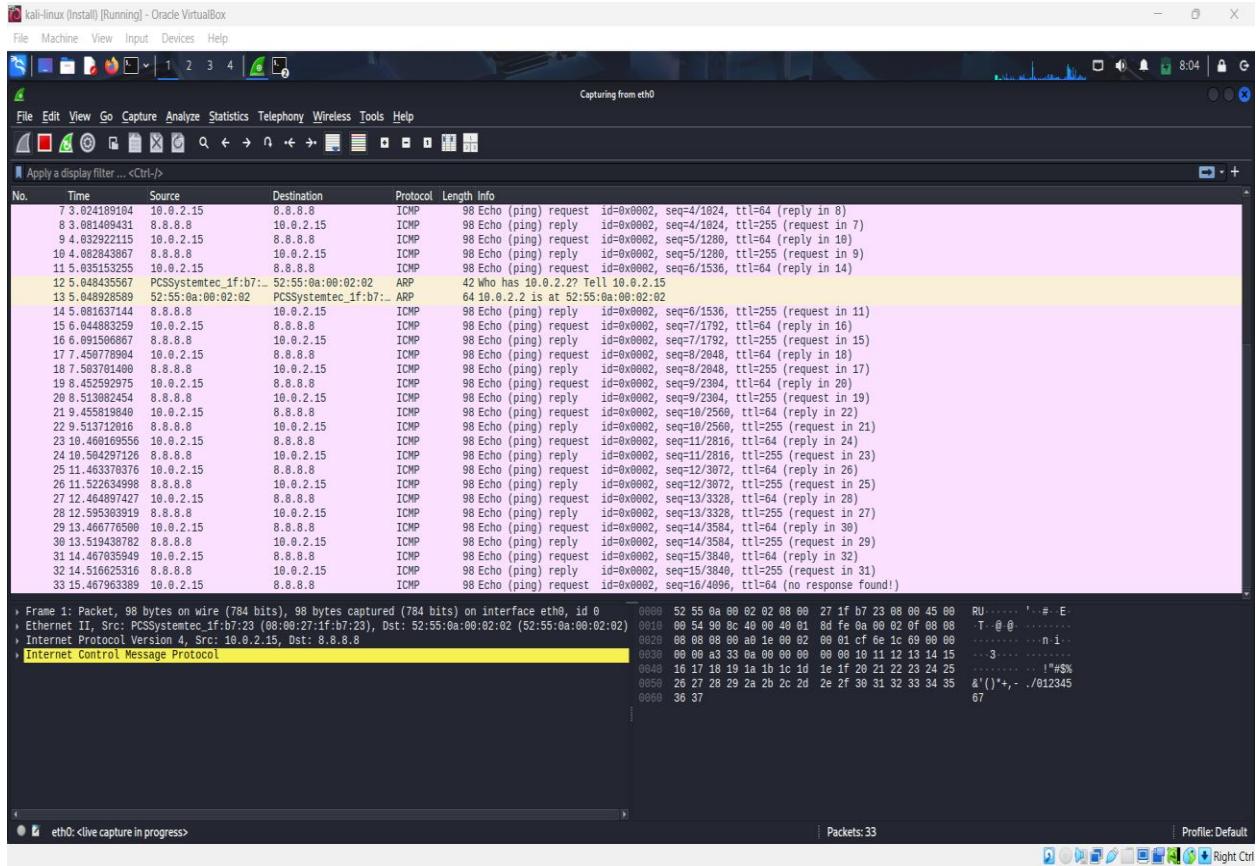
STEP 2 — Start Wireshark

```
sudo wireshark
```

STEP 3 — Start Packet Capture

Inside Wireshark:

1. Select your active interface (likely **eth0** on VM)
2. Click **Blue Shark Icon (Start Capture)**
3. Let it capture for 30 seconds



STEP 4 — Apply Filters

Use filters:

Filter

dns

http

icmp

tcp

udp

Purpose

Only show DNS packets

Only show HTTP packets

Only ping traffic

Only TCP

Only UDP

kali-linux (install) [Running] - Oracle VirtualBox

File Machine View Input Devices Help

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Capturing from eth0

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dns

No.	Time	Source	Destination	Protocol	Length	Info
282	144.846279882	10.0.2.15	10.0.2.3	DNS	88	Standard query 0x8f62 A contile.services.mozilla.com
293	144.846458984	10.0.2.15	10.0.2.3	DNS	88	Standard query 0x3a6c AAAA contile.services.mozilla.com
294	144.961939544	10.0.2.15	10.0.2.3	DNS	79	Standard query 0x9e97 A spocs.getpocket.com
295	144.969864407	10.0.2.15	10.0.2.3	DNS	79	Standard query 0x8ea9 AAAA spocs.getpocket.com
296	145.289468614	10.0.2.15	10.0.2.3	DNS	95	Standard query 0x7567 A content-signature-2.cdn.mozilla.net
297	145.289697663	10.0.2.15	10.0.2.3	DNS	95	Standard query 0x6161 AAAA content-signature-2.cdn.mozilla.net
300	145.928653313	10.0.2.15	10.0.2.3	DNS	87	Standard query 0x6c2 A safebrowsing.googleapis.com
301	145.928942686	10.0.2.15	10.0.2.3	DNS	87	Standard query 0x9c0 AAAA safebrowsing.googleapis.com
302	146.155889588	10.0.2.15	10.0.2.3	DNS	85	Standard query 0x9d1 A push.services.mozilla.com
303	146.156062258	10.0.2.15	10.0.2.3	DNS	85	Standard query 0xf5d4 AAAA push.services.mozilla.com
304	146.365995772	10.0.2.15	10.0.2.3	DNS	97	Standard query 0x827 A firefox.settings.services.mozilla.com
305	146.366139639	10.0.2.15	10.0.2.3	DNS	97	Standard query 0xc825 AAAA firefox.settings.services.mozilla.com
308	146.858088668	10.0.2.15	10.0.2.3	DNS	71	Standard query 0x5309 A example.org
309	146.858098680	10.0.2.15	10.0.2.3	DNS	71	Standard query 0xec9d A example.org
310	146.858828848	10.0.2.15	10.0.2.3	DNS	71	Standard query 0x7a9e AAAA example.org
311	146.868321559	10.0.2.15	10.0.2.3	DNS	73	Standard query 0xc14f A ipv4only.arpa
312	146.868665991	10.0.2.15	10.0.2.3	DNS	73	Standard query 0xc34e AAAA ipv4only.arpa
313	146.868864584	10.0.2.15	10.0.2.3	DNS	84	Standard query 0x7429 A detectportal.firefox.com
314	146.868866196	10.0.2.15	10.0.2.3	DNS	84	Standard query 0x8e2a AAAA detectportal.firefox.com
315	146.86399589	10.0.2.15	10.0.2.3	DNS	84	Standard query 0x870f A detectportal.firefox.com
322	149.857760076	fd17:625c:f037:2:a9...	fd17:625c:f037:2:1:3	DNS	108	Standard query 0x8f62 A contile.services.mozilla.com
323	149.858274880	fd17:625c:f037:2:a9...	fd17:625c:f037:2:1:3	DNS	108	Standard query 0x3a6c AAAA contile.services.mozilla.com
324	149.893170355	fd17:625c:f037:2:1:3	fd17:625c:f037:2:a9...	DNS	124	Standard query response 0x8f62 A contile.services.mozilla.com A 34.36.137.263
325	149.908896330	fd17:625c:f037:2:1:3	fd17:625c:f037:2:a9...	DNS	136	Standard query response 0x3a6c AAAA contile.services.mozilla.com AAAA 64:ff9b::2224:89cb
333	149.982863863	fd17:625c:f037:2:a9...	fd17:625c:f037:2:1:3	DNS	99	Standard query 0x8e97 A spocs.getpocket.com
334	149.982863673	fd17:625c:f037:2:a9...	fd17:625c:f037:2:1:3	DNS	99	Standard query 0x8ea9 AAAA spocs.getpocket.com
335	150.015288893	fd17:625c:f037:2:1:3	fd17:625c:f037:2:a9...	DNS	168	Standard query response 0x9e97 A spocs.getpocket.com CNAME mc.prod.ads.prod.webservices.mozgcp.net A 34.36.137.263
336	150.015298441	fd17:625c:f037:2:1:3	fd17:625c:f037:2:a9...	DNS	180	Standard query response 0x8ea9 AAAA spocs.getpocket.com CNAME mc.prod.ads.prod.webservices.mozgcp.net AAAA 64:ff9b::2224:89cb

Frame 292: Packet, 88 bytes on wire (704 bits), 88 bytes captured (704 bits) on interface eth0, id 0

Ethernet II, Src: PCSSystemtec_1f:b7:23 (08:00:27:1f:b7:23), Dst: 52:55:0a:00:02:03 (52:55:0a:00:02:03)

Internet Protocol Version 4, Src: 10.0.2.15, Dst: 10.0.2.3

User Datagram Protocol, Src Port: 49463, Dst Port: 53

Domain Name System (query)

0000 52 55 0a 00 02 03 08 09 27 1f b7 23 08 06 05 00 RU.....E

0010 00 4a 72 47 40 00 40 11 00 4a 00 00 02 0f 0a 00 Jrg00...J.....

0020 02 03 c1 37 00 35 00 36 10 09 8f 62 01 00 00 01756Yb....

0030 00 00 00 00 00 00 07 63 6f 6e 74 69 6c 65 08 73c ontile s

0040 65 72 76 09 63 65 73 07 6d 6f 7a 69 6c 6c 61 83ervices mozilla

0050 63 6f 6d 00 00 01 00 01 com.....

wireshark_eth0K6A8F3.pcapng

Packets: 3181 - Displayed: 156 (4.9%)

Profile: Default

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File Machine View Input Devices Help

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Capturing from eth0

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arp

No.	Time	Source	Destination	Protocol	Length	Info
12	5.048435567	PCSSystemtec_1f:b7:23	52:55:0a:00:02:02	ARP	42	Who has 10.0.2.2? Tell 10.0.2.15
13	5.048928589	52:55:0a:00:02:02	PCSSystemtec_1f:b7:23	ARP	64	10.0.2.2 is at 52:55:0a:00:02:02
110	53.944647862	PCSSystemtec_1f:b7:23	52:55:0a:00:02:02	ARP	42	Who has 10.0.2.2? Tell 10.0.2.15
111	53.94565884	52:55:0a:00:02:02	PCSSystemtec_1f:b7:23	ARP	64	10.0.2.2 is at 52:55:0a:00:02:02

Frame 12: Packet, 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface eth0, id 0

Ethernet II, Src: PCSSystemtec_1f:b7:23 (08:00:27:1f:b7:23), Dst: 52:55:0a:00:02:02 (52:55:0a:00:02:02)

Address Resolution Protocol (request)

0000 52 55 0a 00 02 02 08 00 27 1f b7 23 08 06 00 01 RU.....

0010 08 00 06 04 00 01 00 00 27 1f b7 23 0a 00 02 0fJ.....

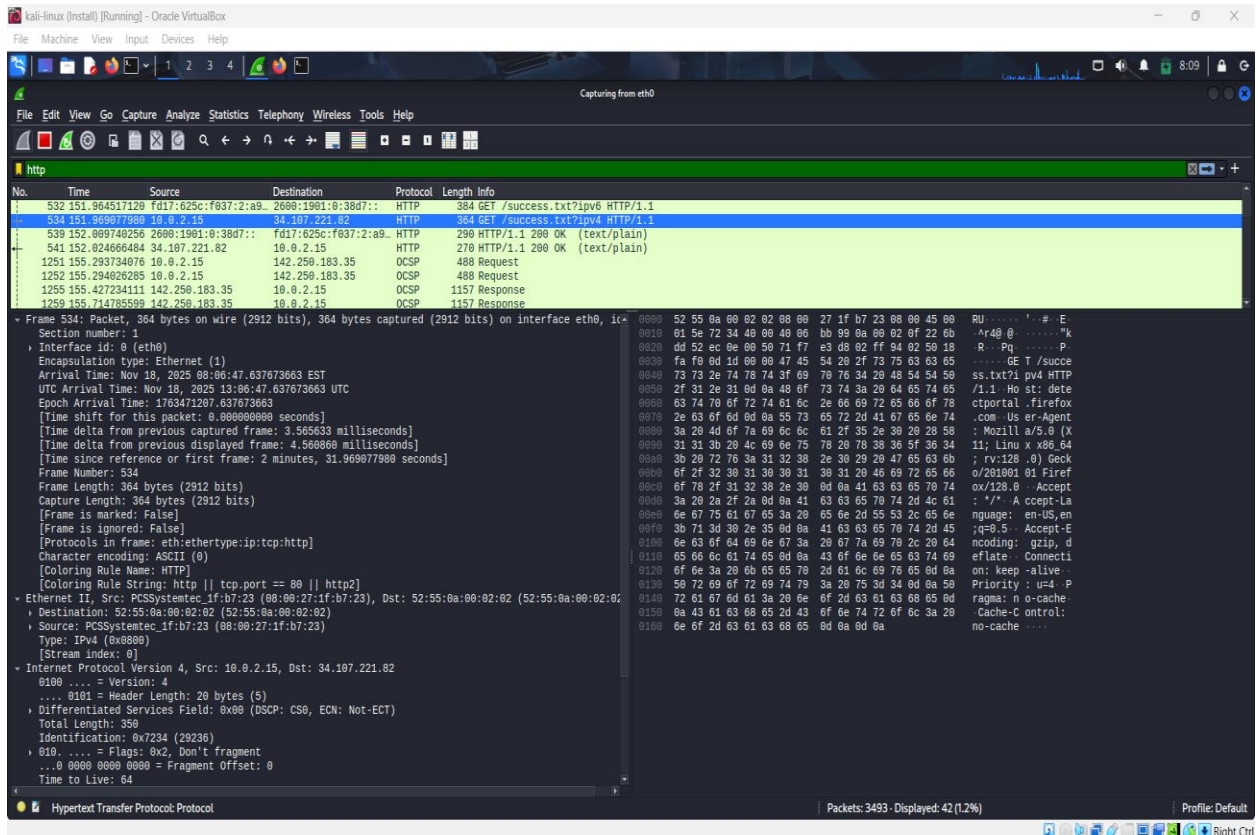
0020 00 00 00 00 00 0a 00 02 02a.....

Address Resolution Protocol: Protocol

Packets: 163 - Displayed: 4 (2.5%)

Profile: Default

ping google.com



STEP 6 — Stop Capture

Click the **red square** (Stop button).

Task 10: Nmap Scans

STEP 1 — Install Nmap

`sudo apt install nmap -y`

STEP 2 — Ping Scan (Live Hosts)

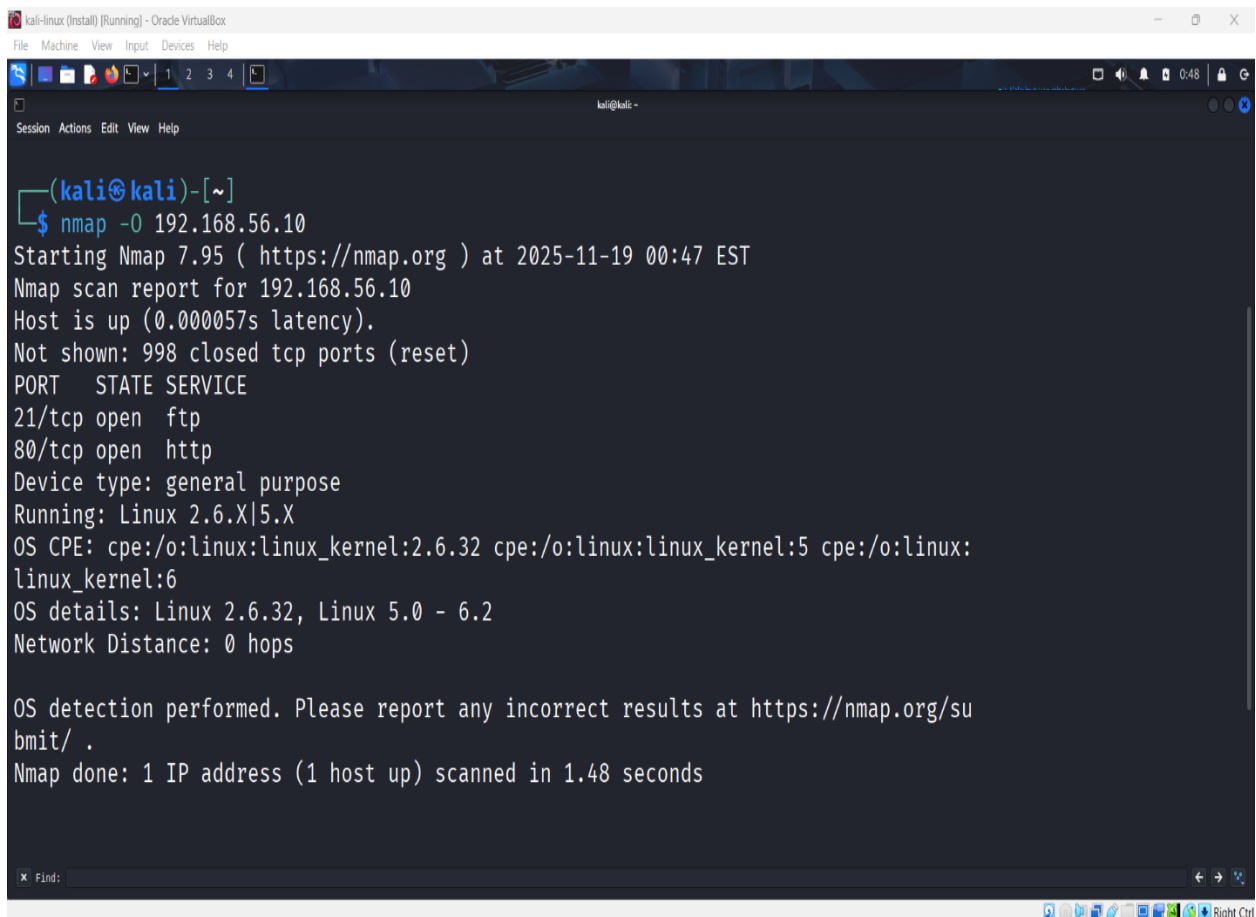
```
nmap -sn <target_ip>
```

STEP 3 — OS Detection Scan

Target any live IP:

```
nmap -O <target_ip>
```

OUTPUT:



```
kali-linux (Install) [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
kali@kali -
Session Actions Edit View Help
(kali@kali)-[~]
$ nmap -O 192.168.56.10
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-19 00:47 EST
Nmap scan report for 192.168.56.10
Host is up (0.000057s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
80/tcp    open  http
Device type: general purpose
Running: Linux 2.6.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:2.6.32 cpe:/o:linux:linux_kernel:5 cpe:/o:linux:linux_kernel:6
OS details: Linux 2.6.32, Linux 5.0 - 6.2
Network Distance: 0 hops

OS detection performed. Please report any incorrect results at https://nmap.org/su
bmit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.48 seconds
```

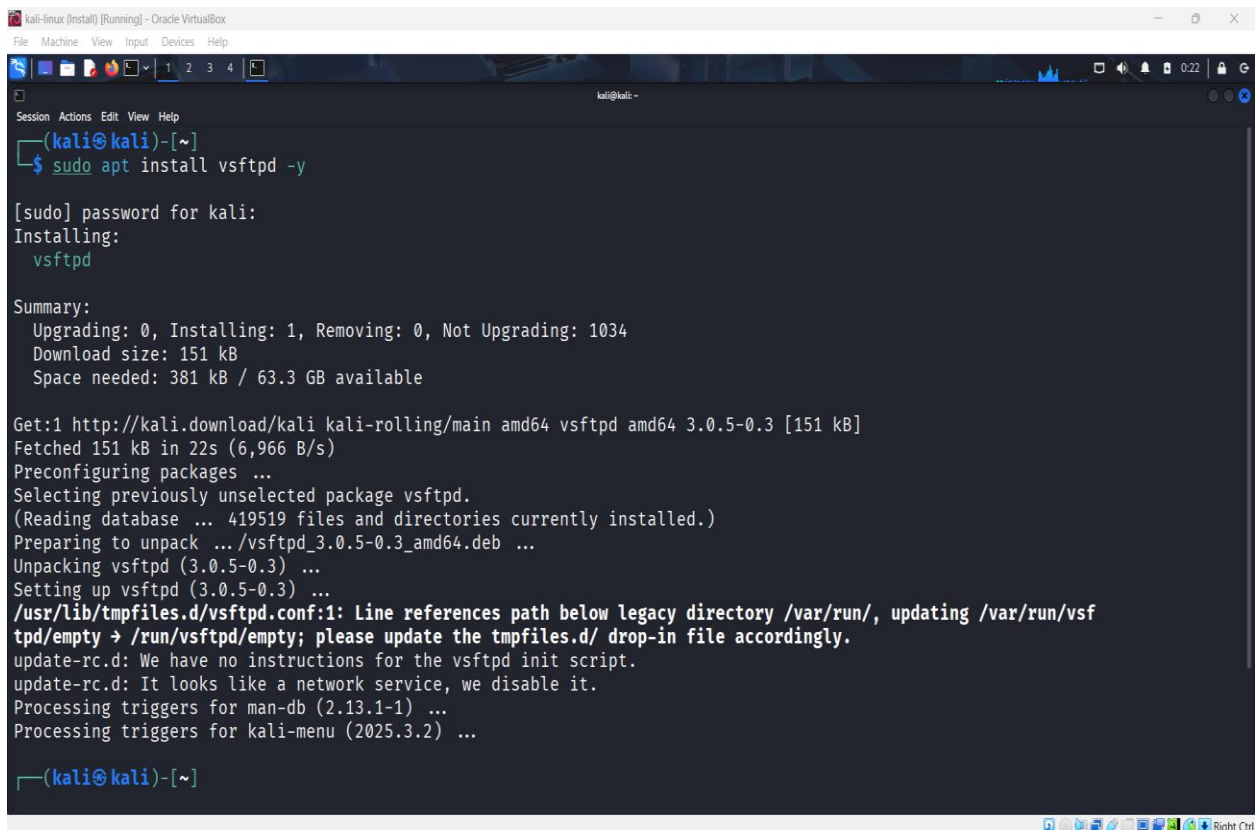


```
(kali@kali)-[~]  
$ nmap -sn 192.168.56.10/24  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-19 00:49 EST  
Nmap scan report for 192.168.56.10  
Host is up.  
Nmap done: 256 IP addresses (1 host up) scanned in 10.76 seconds
```

Task 11: Setup File Transfer (FTP/SFTP)

STEP 1 — Install

`sudo apt install vsftpd -y`



```
kali-linux (Install) [Running] - Oracle VM VirtualBox  
File Machine View Input Devices Help  
kali@kali: ~  
$ sudo apt install vsftpd -y  
[sudo] password for kali:  
Installing:  
vsftpd  
Summary:  
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1034  
  Download size: 151 kB  
  Space needed: 381 kB / 63.3 GB available  
Get:1 http://kali.download/kali kali-rolling/main amd64 vsftpd amd64 3.0.5-0.3 [151 kB]  
Fetched 151 kB in 22s (6,966 B/s)  
Preconfiguring packages ...  
Selecting previously unselected package vsftpd.  
(Reading database ... 419519 files and directories currently installed.)  
Preparing to unpack .../vsftpd_3.0.5-0.3_amd64.deb ...  
Unpacking vsftpd (3.0.5-0.3) ...  
Setting up vsftpd (3.0.5-0.3) ...  
/usr/lib/tmpfiles.d/vsftpd.conf:1: Line references path below legacy directory /var/run/, updating /var/run/vsftpd/empty → /run/vsftpd/empty; please update the tmpfiles.d/ drop-in file accordingly.  
update-rc.d: We have no instructions for the vsftpd init script.  
update-rc.d: It looks like a network service, we disable it.  
Processing triggers for man-db (2.13.1-1) ...  
Processing triggers for kali-menu (2025.3.2) ...  
kali@kali: ~
```

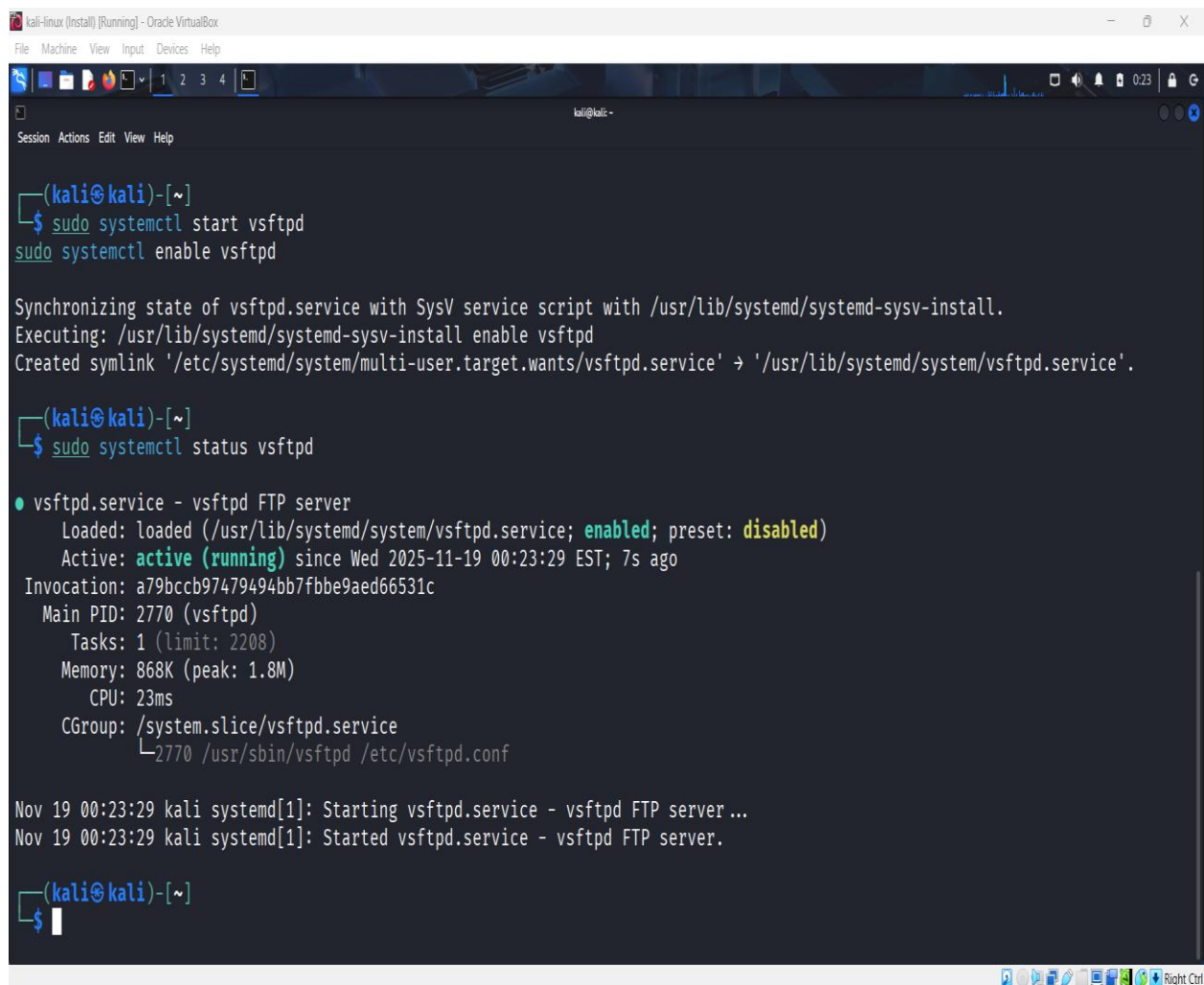
STEP 2 — Start Service

```
sudo systemctl start vsftpd
```

```
sudo systemctl enable vsftpd
```

STEP 3 — Check Status

```
sudo systemctl status vsftpd
```



```
kali-linux (Install) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

(kali@kali)-[~]
$ sudo systemctl start vsftpd
sudo systemctl enable vsftpd

Synchronizing state of vsftpd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable vsftpd
Created symlink '/etc/systemd/system/multi-user.target.wants/vsftpd.service' -> '/usr/lib/systemd/system/vsftpd.service'.

(kali@kali)-[~]
$ sudo systemctl status vsftpd

● vsftpd.service - vsftpd FTP server
   Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; enabled; preset: disabled)
   Active: active (running) since Wed 2025-11-19 00:23:29 EST; 7s ago
  Invocation: a79bccb97479494bb7fbb9aed66531c
    Main PID: 2770 (vsftpd)
      Tasks: 1 (limit: 2208)
     Memory: 868K (peak: 1.8M)
        CPU: 23ms
    CGroup: /system.slice/vsftpd.service
            └─2770 /usr/sbin/vsftpd /etc/vsftpd.conf

Nov 19 00:23:29 kali systemd[1]: Starting vsftpd.service - vsftpd FTP server ...
Nov 19 00:23:29 kali systemd[1]: Started vsftpd.service - vsftpd FTP server.

(kali@kali)-[~]
$
```


STEP 4 — Enable Upload

Edit configuration:

```
sudo nano /etc/vsftpd.conf
```

Find/uncomment/add:

```
write_enable=YES
```

```
local_umask=022
```

Restart service:

```
sudo systemctl restart vsftpd
```

STEP 5 — FTP Test From Kali

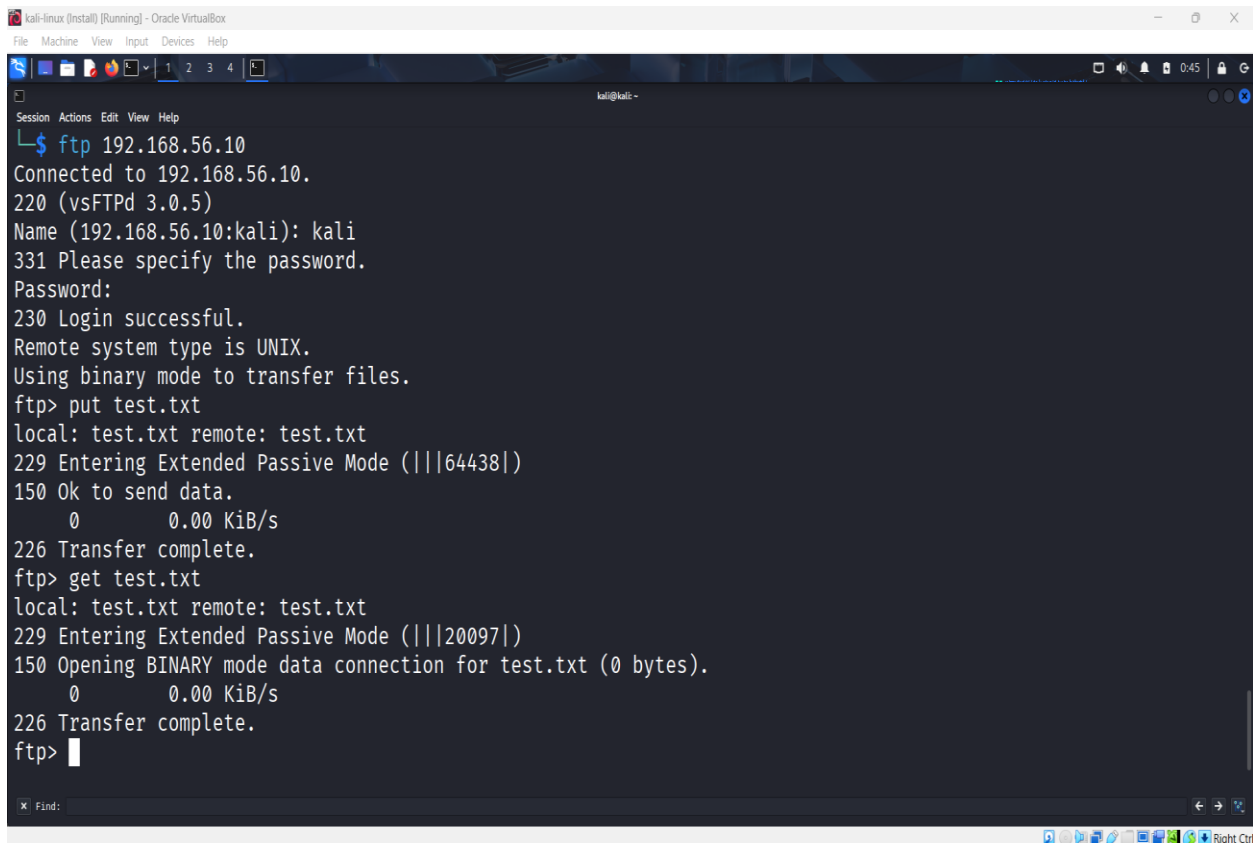
```
ftp <your_ip>
```

When connected, try upload:

```
put test.txt
```

Download:

```
get test.tx
```



```
kali-linux (Install) [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

kali@kali:~$ ftp 192.168.56.10
Connected to 192.168.56.10.
220 (vsFTPd 3.0.5)
Name (192.168.56.10:kali): kali
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> put test.txt
local: test.txt remote: test.txt
229 Entering Extended Passive Mode (|||64438|)
150 Ok to send data.
0 0.00 KiB/s
226 Transfer complete.
ftp> get test.txt
local: test.txt remote: test.txt
229 Entering Extended Passive Mode (|||20097|)
150 Opening BINARY mode data connection for test.txt (0 bytes).
0 0.00 KiB/s
226 Transfer complete.
ftp> 
```

TASK 12: Network Topology Diagram

Client— 10.0.2.15 & 192.168.56.101

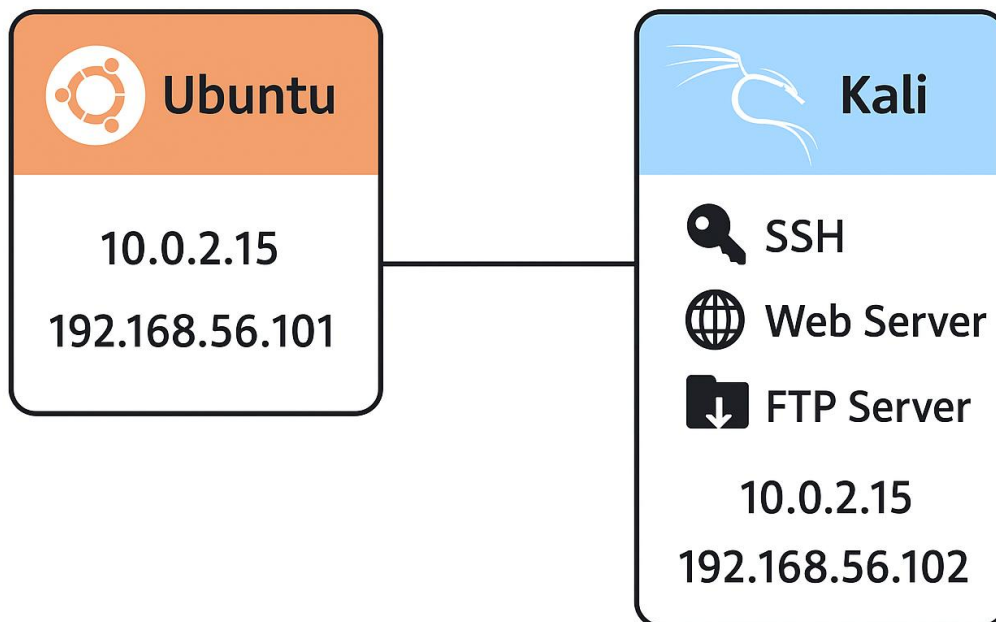
|

| — HTTP, FTP, SSH, Ping

|

Web Server, FTP Server , SSH (Kali)

10.0.2.15 & 192.168.56.102



TASK 13: LAN Setup (3 VMs)

STEP 1 — Setup VirtualBox Network

Use NAT Network or Host-Only Network.

STEP 2 — Assign IPs

On each VM run:

`ip a`

VM 1: Kali (Client)

Already installed tools:

- nmap
- curl
- ftp
- ssh

```
Session Actions Edit View Help
(kali@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:1f:b7:23 brd ff:ff:ff:ff:ff:ff
   inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 81075sec preferred_lft 81075sec
   inet6 fd17:625c:f037:2:a9f1:899c:8ef3:741d/64 scope global dynamic noprefixroute
       valid_lft 86342sec preferred_lft 14342sec
   inet6 fe80::2d15:152:6bd5:553f/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
   link/ether 08:00:27:8a:05:ae brd ff:ff:ff:ff:ff:ff
   inet 192.168.56.102/24 brd 192.168.56.255 scope global dynamic noprefixroute eth1
       valid_lft 374sec preferred_lft 374sec
   inet6 fe80::c58b:8b2c:54eb:dd60/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

VM 2: Web Server (Apache)

```

root@Ubuntu:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:57:b4:67 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 82776sec preferred_lft 82776sec
    inet6 fd17:625c:f037:2:cadca166:8627:2674/64 scope global temporary dynamic
        valid_lft 85934sec preferred_lft 13934sec
    inet6 fd17:625c:f037:2:a00:27ff:fe57:b467/64 scope global dynamic mngtmpaddr
        valid_lft 85934sec preferred_lft 13934sec
    inet6 fe80::a00:27ff:fe57:b467/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:ce:78:3a brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.101/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s8
        valid_lft 575sec preferred_lft 575sec
    inet6 fe80::2ea5:c28a:1019:7da3/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

```

Install:

`sudo apt install apache2 -y`

Check:

`sudo systemctl status apache2`

```

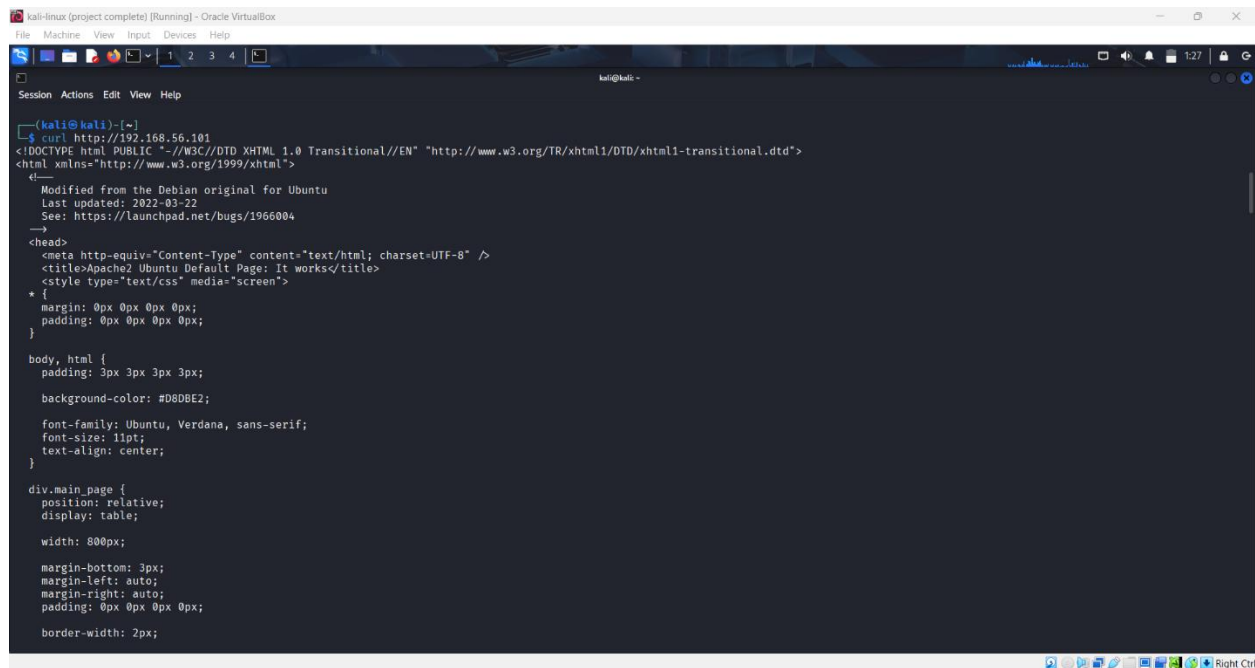
root@Ubuntu:~# sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Sat 2025-11-22 17:39:14 UTC; 1h 8min ago
     Docs: https://httpd.apache.org/docs/2.4/
   Process: 1330 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
  Main PID: 1706 (apache2)
    Tasks: 7 (limit: 3468)
   Memory: 23.4M (peak: 23.6M)
      CPU: 1.510s
   CGroup: /system.slice/apache2.service
           └─1706 /usr/sbin/apache2 -k start
             └─2068 /usr/sbin/apache2 -k start
               └─2069 /usr/sbin/apache2 -k start
                 └─2070 /usr/sbin/apache2 -k start
                   └─2071 /usr/sbin/apache2 -k start
                     └─2072 /usr/sbin/apache2 -k start
                       └─4591 /usr/sbin/apache2 -k start

Nov 22 17:39:01 Ubuntu systemd[1]: Starting apache2.service - The Apache HTTP Server...
Nov 22 17:39:14 Ubuntu apachectl[1372]: apache2: Could not reliably determine the server's fully qualified domain name, using 10.0.2.15. Set the 'ServerName' directive global
Nov 22 17:39:14 Ubuntu systemd[1]: Started apache2.service - The Apache HTTP Server.
[lines 1-21/21 (END)]

```

Test:

`curl http://<web_server_ip>`



```
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File Machine View Input Devices Help

(kali@kali)-[~]
$ curl http://192.168.56.101
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<!--
  Modified from the Debian original for Ubuntu
  Last updated: 2022-03-22
  See: https://launchpad.net/bugs/1966004
-->
<head>
<meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
<title>Apache2 Ubuntu Default Page: It works</title>
<style type="text/css" media="screen">
* {
  margin: 0px 0px 0px 0px;
  padding: 0px 0px 0px 0px;
}

body, html {
  padding: 3px 3px 3px 3px;

  background-color: #D8DBE2;

  font-family: Ubuntu, Verdana, sans-serif;
  font-size: 11pt;
  text-align: center;
}

div.main_page {
  position: relative;
  display: table;

  width: 800px;

  margin-bottom: 3px;
  margin-left: auto;
  margin-right: auto;
  padding: 0px 0px 0px 0px;

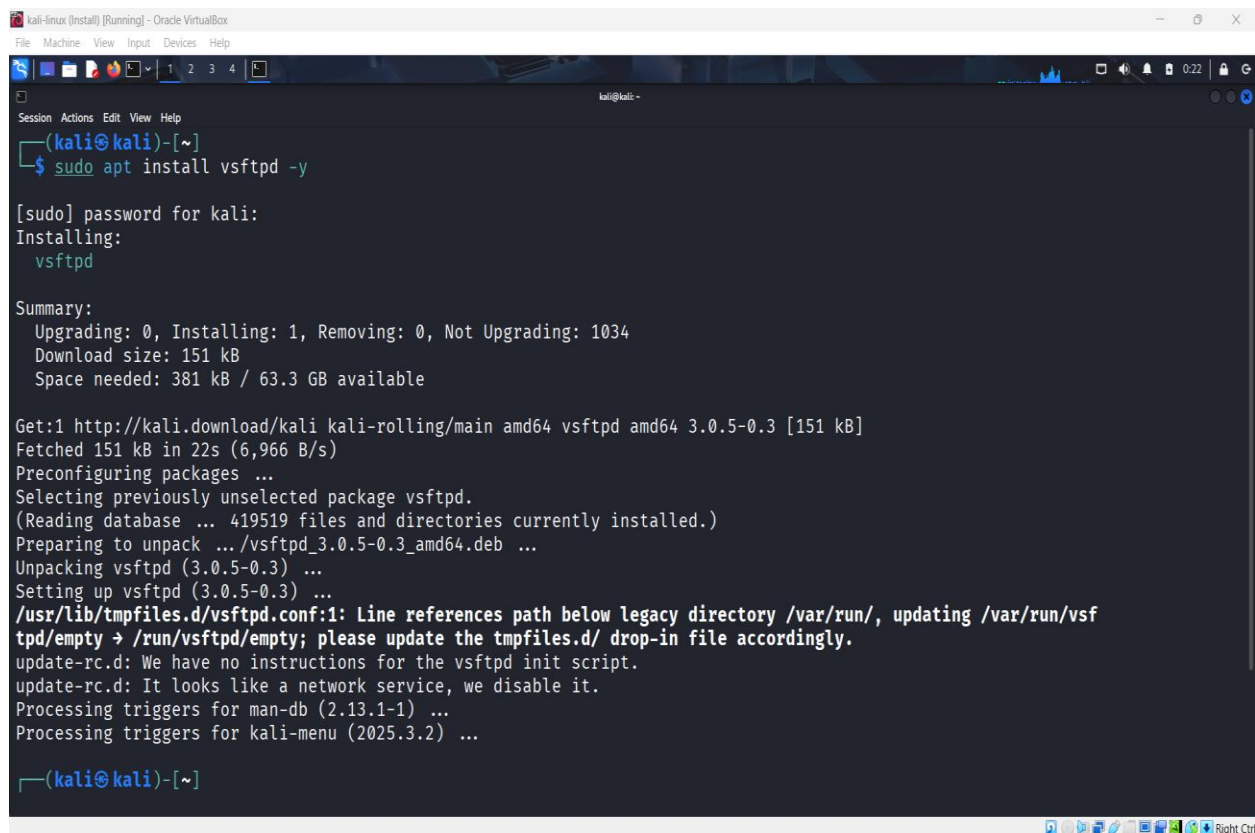
  border-width: 2px;
```

VM 3: FTP Server (vsftpd)

```
(kali@kali)-[~]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1
    000
    link/ether 08:00:27:f9:e8:83 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 86382sec preferred_lft 86382sec
    inet6 fd17:625c:f037:2:6d33:355d:e7b1:a/64 scope global dynamic noprefixroute
        valid_lft 86383sec preferred_lft 14383sec
    inet6 fe80::26a4:2495:e7fe:81d3/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1
    000
    link/ether 08:00:27:db:ad:20 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.105/24 brd 192.168.56.255 scope global dynamic noprefixroute eth1
        valid_lft 582sec preferred_lft 582sec
    inet6 fe80::ffde:e388:15d2:4789/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Install

`sudo apt install vsftpd -y`

A screenshot of a Kali Linux terminal window. The window title is "kali-linux (install) [Running] - Oracle VM VirtualBox". The terminal shows the command `sudo apt install vsftpd -y` being executed. It prompts for a password, then shows the installation progress for vsftpd. The summary indicates that 1 package is being installed, with a download size of 151 kB and a space requirement of 381 kB. The terminal output shows the package being fetched, unpacked, and configured. A warning message is displayed: `/usr/lib/tmpfiles.d/vsftpd.conf:1: Line references path below legacy directory /var/run/, updating /var/run/vsftpd/empty → /run/vsftpd/empty; please update the tmpfiles.d/ drop-in file accordingly.` The terminal ends with the prompt `(kali@kali)-[~]`.

```
kali@kali~$ sudo apt install vsftpd -y
[sudo] password for kali:
Installing:
  vsftpd

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1034
  Download size: 151 kB
  Space needed: 381 kB / 63.3 GB available

Get:1 http://kali.download/kali kali-rolling/main amd64 vsftpd amd64 3.0.5-0.3 [151 kB]
Fetched 151 kB in 22s (6,966 B/s)
Preconfiguring packages ...
Selecting previously unselected package vsftpd.
(Reading database ... 419519 files and directories currently installed.)
Preparing to unpack .../vsftpd_3.0.5-0.3_amd64.deb ...
Unpacking vsftpd (3.0.5-0.3) ...
Setting up vsftpd (3.0.5-0.3) ...
/usr/lib/tmpfiles.d/vsftpd.conf:1: Line references path below legacy directory /var/run/, updating /var/run/vsftpd/empty → /run/vsftpd/empty; please update the tmpfiles.d/ drop-in file accordingly.
update-rc.d: We have no instructions for the vsftpd init script.
update-rc.d: It looks like a network service, we disable it.
Processing triggers for man-db (2.13.1-1) ...
Processing triggers for kali-menu (2025.3.2) ...

(kali@kali)-[~]
```

Start Service

```
sudo systemctl start vsftpd
```

```
sudo systemctl enable vsftpd
```

Check Status

```
sudo systemctl status vsftpd
```

```
(kali㉿kali)-[~]
└─$ sudo systemctl status vsftpd
● vsftpd.service - vsftpd FTP server
   Loaded: loaded (/usr/lib/systemd/system/vsftpd.service; enabled; preset: disabled)
   Active: active (running) since Sun 2025-11-23 01:53:20 EST; 12s ago
     Invocation: e1aab7f9154b4b0988c3419a3f905454
    Main PID: 5179 (vsftpd)
      Tasks: 1 (limit: 2208)
     Memory: 956K (peak: 1.7M)
        CPU: 32ms
    CGroup: /system.slice/vsftpd.service
            └─5179 /usr/sbin/vsftpd /etc/vsftpd.conf

Nov 23 01:53:20 kali systemd[1]: Starting vsftpd.service - vsftpd FTP server...
Nov 23 01:53:20 kali systemd[1]: Started vsftpd.service - vsftpd FTP server.
```

STEP 3 — Connectivity Test

Ping Web Server:

ping <web_server_ip>

```
(kali㉿kali)-[~]
└─$ ping 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data.
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=2.19 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=1.31 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=1.76 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=2.01 ms
64 bytes from 192.168.56.101: icmp_seq=5 ttl=64 time=1.86 ms
64 bytes from 192.168.56.101: icmp_seq=6 ttl=64 time=1.83 ms
64 bytes from 192.168.56.101: icmp_seq=7 ttl=64 time=2.31 ms
64 bytes from 192.168.56.101: icmp_seq=8 ttl=64 time=1.86 ms
64 bytes from 192.168.56.101: icmp_seq=9 ttl=64 time=2.32 ms
64 bytes from 192.168.56.101: icmp_seq=10 ttl=64 time=2.13 ms
64 bytes from 192.168.56.101: icmp_seq=11 ttl=64 time=2.43 ms
64 bytes from 192.168.56.101: icmp_seq=12 ttl=64 time=1.68 ms
64 bytes from 192.168.56.101: icmp_seq=13 ttl=64 time=1.72 ms
64 bytes from 192.168.56.101: icmp_seq=14 ttl=64 time=2.02 ms
64 bytes from 192.168.56.101: icmp_seq=15 ttl=64 time=1.00 ms
64 bytes from 192.168.56.101: icmp_seq=16 ttl=64 time=1.09 ms
64 bytes from 192.168.56.101: icmp_seq=17 ttl=64 time=4.09 ms
64 bytes from 192.168.56.101: icmp_seq=18 ttl=64 time=2.25 ms
64 bytes from 192.168.56.101: icmp_seq=19 ttl=64 time=1.44 ms
64 bytes from 192.168.56.101: icmp_seq=20 ttl=64 time=1.99 ms
```

Ping FTP Server:

ping <ftp_server_ip>

```
(kali㉿kali)-[~]  
$ ping 192.168.56.105  
PING 192.168.56.105 (192.168.56.105) 56(84) bytes of data.  
64 bytes from 192.168.56.105: icmp_seq=1 ttl=64 time=0.038 ms  
64 bytes from 192.168.56.105: icmp_seq=2 ttl=64 time=0.057 ms  
64 bytes from 192.168.56.105: icmp_seq=3 ttl=64 time=0.042 ms  
64 bytes from 192.168.56.105: icmp_seq=4 ttl=64 time=0.045 ms  
64 bytes from 192.168.56.105: icmp_seq=5 ttl=64 time=0.041 ms  
64 bytes from 192.168.56.105: icmp_seq=6 ttl=64 time=0.046 ms  
64 bytes from 192.168.56.105: icmp_seq=7 ttl=64 time=0.067 ms  
64 bytes from 192.168.56.105: icmp_seq=8 ttl=64 time=0.035 ms  
64 bytes from 192.168.56.105: icmp_seq=9 ttl=64 time=0.054 ms  
64 bytes from 192.168.56.105: icmp_seq=10 ttl=64 time=0.049 ms  
64 bytes from 192.168.56.105: icmp_seq=11 ttl=64 time=0.037 ms  
64 bytes from 192.168.56.105: icmp_seq=12 ttl=64 time=0.038 ms  
64 bytes from 192.168.56.105: icmp_seq=13 ttl=64 time=0.042 ms  
64 bytes from 192.168.56.105: icmp_seq=14 ttl=64 time=0.058 ms  
64 bytes from 192.168.56.105: icmp_seq=15 ttl=64 time=0.059 ms  
64 bytes from 192.168.56.105: icmp_seq=16 ttl=64 time=0.047 ms  
64 bytes from 192.168.56.105: icmp_seq=17 ttl=64 time=0.044 ms  
64 bytes from 192.168.56.105: icmp_seq=18 ttl=64 time=0.064 ms  
64 bytes from 192.168.56.105: icmp_seq=19 ttl=64 time=0.049 ms  
64 bytes from 192.168.56.105: icmp_seq=20 ttl=64 time=0.054 ms  
64 bytes from 192.168.56.105: icmp_seq=21 ttl=64 time=0.048 ms  
64 bytes from 192.168.56.105: icmp_seq=22 ttl=64 time=0.047 ms  
64 bytes from 192.168.56.105: icmp_seq=23 ttl=64 time=0.043 ms  
64 bytes from 192.168.56.105: icmp_seq=24 ttl=64 time=0.072 ms  
64 bytes from 192.168.56.105: icmp_seq=25 ttl=64 time=0.081 ms
```

TASK 14: Test Web/FTP Using Nmap

Scan Web Server

nmap -sV <web_server_ip>

```
(kali㉿kali)-[~]
$ nmap -sV 192.168.56.101
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-23 01:20 EST
Nmap scan report for 192.168.56.101
Host is up (0.0032s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.5
22/tcp    open  ssh      OpenSSH 9.6p1 Ubuntu 3ubuntu13.14 (Ubuntu Linux; protocol 2.0)
23/tcp    open  telnet?
80/tcp    open  http     Apache httpd 2.4.58 ((Ubuntu))
MAC Address: 08:00:27:CE:78:3A (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 165.67 seconds
```

Scan FTP Server

`nmap -sV <ftp_server_ip>`

```
(kali㉿kali)-[~]
$ nmap -sV 192.168.56.105
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-23 01:58 EST
Nmap scan report for 192.168.56.105
Host is up (0.0018s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.5
80/tcp    open  http     Apache httpd 2.4.65 ((Debian))
MAC Address: 08:00:27:DB:AD:20 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.89 seconds
```

Curl Test

`curl http://<web_server_ip>`

```
kali-linux (project complete) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

(kali@kali)-[~]
└─$ curl http://192.168.56.105
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
  <!--
    Modified from the Debian original for Ubuntu
    Last updated: 2022-03-22
    See: https://launchpad.net/bugs/1966004
  -->
  <head>
    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
    <title>Apache2 Ubuntu Default Page: It works</title>
    <style type="text/css" media="screen">
      * {
        margin: 0px 0px 0px 0px;
        padding: 0px 0px 0px 0px;
      }

      body, html {
        padding: 3px 3px 3px 3px;
        background-color: #D8DBE2;
        font-family: Ubuntu, Verdana, sans-serif;
        font-size: 11pt;
        text-align: center;
      }

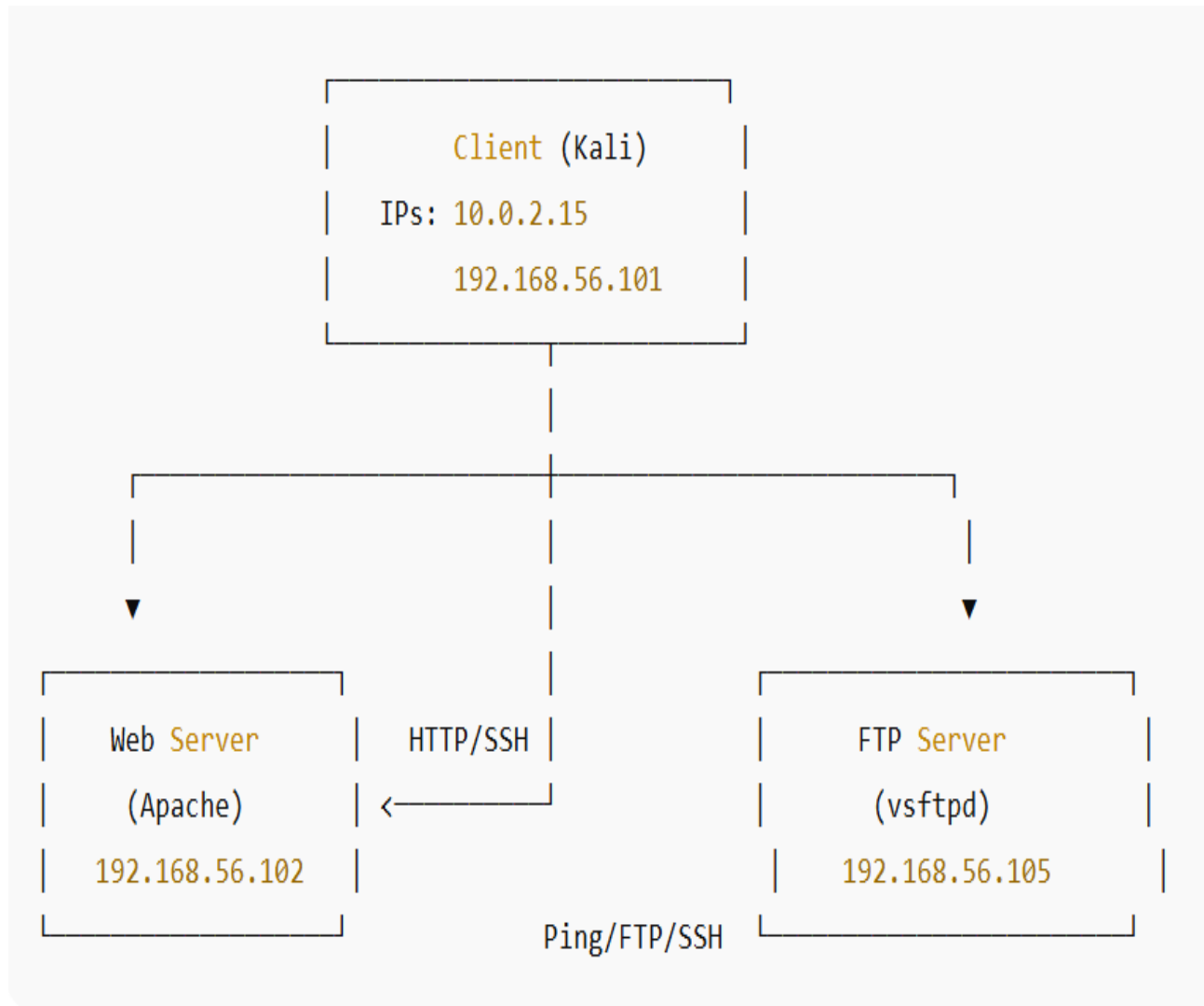
      div.main_page {
        position: relative;
        display: table;
        width: 800px;
        margin-bottom: 3px;
        margin-left: auto;
        margin-right: auto;
        padding: 0px 0px 0px 0px;
        border-width: 2px;
      }
    </style>
  </head>
  <body>
    <div class="main_page">
      <div>
        <h1>It works!</h1>
      </div>
    </div>
  </body>
</html>
```

FTP Test

ftp <ftp_server_ip>

```
(kali@kali)-[~]
└─$ ftp 192.168.56.105
Connected to 192.168.56.105.
220 (vsFTPd 3.0.5)
Name (192.168.56.105:kali): kali
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (|||28465|)
150 Here comes the directory listing.
drwxr-xr-x  2 1000  1000           4096 Nov 09 02:27 Desktop
drwxr-xr-x  2 1000  1000           4096 Nov 09 02:27 Documents
drwxr-xr-x  2 1000  1000           4096 Nov 09 02:27 Downloads
drwxr-xr-x  2 1000  1000           4096 Nov 09 02:27 Music
drwxr-xr-x  2 1000  1000           4096 Nov 09 02:27 Pictures
drwxr-xr-x  2 1000  1000           4096 Nov 09 02:27 Public
drwxr-xr-x  2 1000  1000           4096 Nov 09 02:27 Templates
drwxr-xr-x  2 1000  1000           4096 Nov 09 02:27 Videos
226 Directory send OK.
ftp> █
```

Network Topology Diagram



Conclusion

During Week 3 and Week 4, I gained practical, hands-on experience with core networking tools and real-world security workflows. I successfully captured and analyzed live network traffic using Wireshark, performed host discovery and OS fingerprinting with Nmap, configured an operational FTP server using vsftpd, and designed a functional network topology.

By Week 4, I implemented a complete mini LAN project containing a Web Server, FTP Server, and Kali Client. I validated connectivity, tested HTTP and FTP services, and performed detailed port and service enumeration. This process helped reinforce my understanding of network communication, service behavior, and potential security implications.

Overall, these tasks strengthened my foundational skills in network analysis, scanning, system configuration, and service deployment. This hands-on exposure prepares me for more advanced cybersecurity activities such as penetration testing, vulnerability assessment, and secure infrastructure design.